



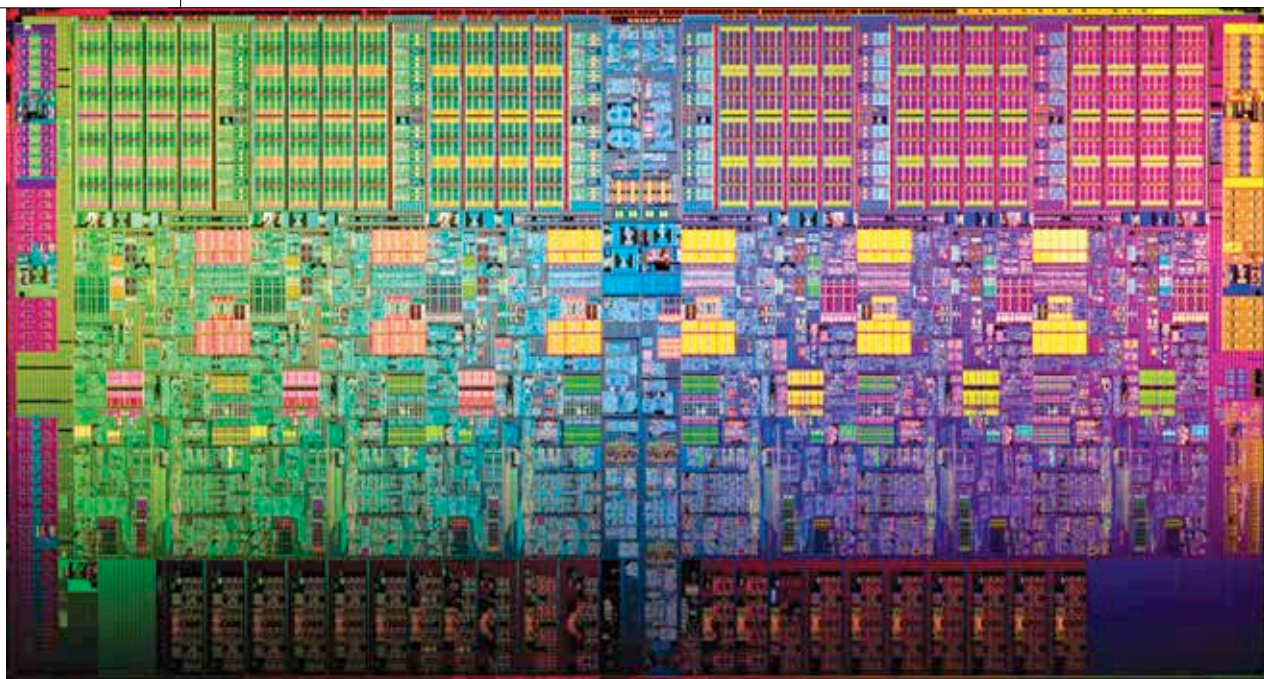
ASUS pop-up, punch hole

New patents reveal that ASUS could be working on phones with pop-up and punch-hole cameras. <https://digit.in/jan19-5>



Uber settles

Uber has offered a tentative settlement to drivers who've pursued arbitration cases over their worker status in the US. <https://digit.in/jan19-6>



IT'S SUNNY IN PORTLAND

Intel's next microarchitecture is a major upheaval over the Skylake architecture that has been in the limelight for way too long.

Mithun Mohandas | mithun@digit.in

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t the recent Intel Architecture Day, the company made some key announcements that finally let most

analysts to believe that the stagnation that has been around since Skylake launched would finally come to an end. The announcements were spread across six domains with the first one being manufacturing process, followed by architecture, memory technology, interconnect, security and finally, software.

STAGNATION OR A 10NM PROCESS THAT WASN'T READY?

Skylake came out in 2015, and every generation of Intel CPUs that were launched since there were based on the Skylake microarchitecture. The only differences in the subsequent 'xxx'lake microarchitectures was the improvement in frequency and the manufacturing process node. Skylake was made on a 14nm process node and had significant improvements in terms of single-threaded performance, reduced power consumption and frequency. Kabylake came in 2016 and brought along with it a slight improvement in frequency and used the 14nm+ process

node. The trend continued with Coffee Lake in 2017 and Coffee Lake Refresh in 2018 as well.

Essentially, all Intel processors from Skylake onwards up until the recent Coffee Lake Refresh were simply minor improvements. Ever since 2004, every two years we've seen a shrink in the processor node. After 14nm in 2014, the 10nm process node was set to follow in 2016 but only Samsung managed to get their fabs to produce silicon on the 10nm process node in that year. Everyone else was struggling with it and after four years of being on 14nm, industry analysts started reporting in late October that Intel had killed off their 10nm plans. Global Foundries had already skipped 10nm and TSMC believed that 10nm would be a short lived one, and thus, wouldn't be worth the effort. Intel, of course, denied all reports regarding killing off its own 10nm plans but did not reveal if they had any silicon ready for the elusive 10nm process node.

It was only at the Intel Architecture Day that they finally showcased working products built on 10nm. The new architecture to use Intel's 10nm is now called Sunny Cove. Given that Intel has demonstrated Sunny Cove, it's safe to say that we will see the next generation of Intel processors utilising